

# The second criterion of congruence of triangles

A side and the two angles on it — the criterion that surveyors use.

LESSON 51–52

2 × 40 MIN

MATEMATIKA 7, §22

S8 5

LESSON CLOCK

15'

25'

25'

20'

5'

|                      |        |
|----------------------|--------|
| The criterion        | 15 min |
| Which side counts    | 25 min |
| In proofs            | 25 min |
| Measuring a distance | 20 min |
| Homework             | 5 min  |

LEARNING OBJECTIVES

- State and apply the ASA criterion.
- Recognise when the given side is between the two angles.
- Use the criterion in proofs involving parallel lines.
- Apply it to measuring an inaccessible distance.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 137–142        |
| Cambridge | Stage 8, pp. 48–56 |
| Workbook  | Exercise 5.1       |

01

## Explanation

### The criterion

If a side and the two angles on it in one triangle equal a side and the two angles on it in another, the triangles are congruent

$$BC = B_1C_1, \angle B = \angle B_1, \angle C = \angle C_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

The third angle is then equal too, since the angles of a triangle add to 180°.

TWO ANGLES AND ANY SIDE ARE ENOUGH

Because the third angle follows from the angle sum, a side and any two angles determine the triangle — whether the side lies between them or not. That is why the criterion is sometimes written ASA and sometimes AAS.

### Which side counts

| GIVEN                          | ENOUGH? | WHY                             |
|--------------------------------|---------|---------------------------------|
| $a, \angle B, \angle C$        | yes     | the side is between them        |
| $a, \angle A, \angle B$        | yes     | $\angle C$ follows from the sum |
| $\angle A, \angle B, \angle C$ | no      | no side fixes the size          |

The middle row is the one that surprises: the side need not be included, because the missing angle is never really missing.

### THREE ANGLES FIX THE SHAPE, NEVER THE SIZE

Any enlargement of a triangle has the same three angles. At least one length is always needed.

## In proofs

**Problem.**  $AB \parallel CD$  and  $AB = CD$ . Prove  $\triangle ABO \equiv \triangle DCO$ , where  $O$  is the intersection of  $AD$  and  $BC$ .

| STATEMENT                            | REASON                              |
|--------------------------------------|-------------------------------------|
| $AB = CD$                            | given                               |
| $\angle ABO = \angle DCO$            | alternate angles, $AB \parallel CD$ |
| $\angle BAO = \angle CDO$            | alternate angles                    |
| $\triangle ABO \equiv \triangle DCO$ | second criterion                    |

### PARALLEL LINES ARE A SUPPLY OF EQUAL ANGLES

Almost every use of the second criterion begins with a pair of parallels. Spotting them turns a hard figure into an easy proof.

## Measuring a distance

To measure the width of a river without crossing it:

1. mark  $A$  on one bank and a landmark  $B$  on the other;
2. walk along the bank to  $C$  and measure  $AC$ ;
3. measure  $\angle BAC$  and  $\angle BCA$ ;

4. construct a triangle with the same side and two angles on level ground, and measure its third side.

By the second criterion the constructed triangle is congruent to the real one, so the measured side **is** the width of the river.

THIS IS WHAT SURVEYING WAS, BEFORE TRIGONOMETRY

A tape and a protractor, a criterion of congruence, and a drawing on level ground. The method is ancient, exact, and still taught for exactly that reason.

## 02 Worked examples

**EXAMPLE 1** Two triangles have a side of 7 with angles  $50^\circ$  and  $60^\circ$  on it. Are they congruent?

- ① A side and the two angles on it.
- ② The second criterion applies.
- ③ The third angle is  $70^\circ$  in both.
- ④ Congruent.

ANSWER

Yes, by ASA

**EXAMPLE 2** Two triangles have a side of 7, an angle of  $50^\circ$  on it and an angle of  $70^\circ$  opposite it. Congruent?

- 1 The third angle is  $180 - 50 - 70 = 60^\circ$ .
- 2 So all three angles are known in both.
- 3 With the side of 7 in the same position, ASA applies.
- 4 Congruent.  
The side need not be included.

ANSWER

Yes

**EXAMPLE 3**  $AB \parallel CD$  and  $AB = CD$ . Prove that  $AD$  and  $BC$  bisect each other.

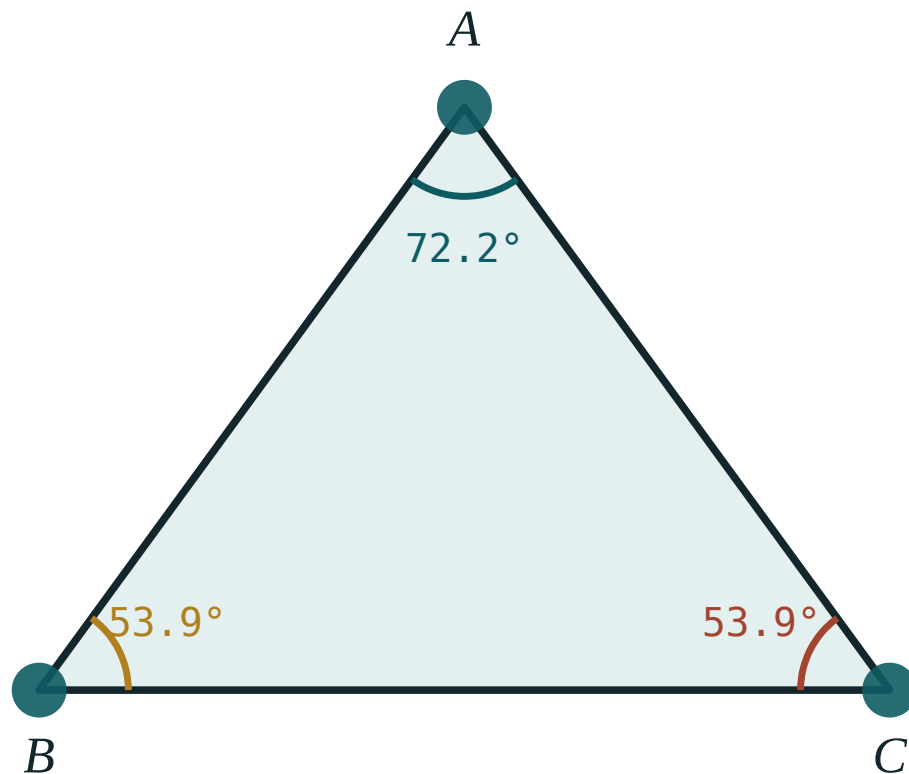
- 1 Alternate angles give  $\angle ABO = \angle DCO$  and  $\angle BAO = \angle CDO$ .
- 2 With  $AB = CD$ , the second criterion gives  $\triangle ABO \equiv \triangle DCO$ .
- 3 CPCT:  $AO = DO$  and  $BO = CO$ .
- 4 So each diagonal is bisected.

ANSWER

Proved

### 03 Interactive model

Measure the width of the corridor from inside the classroom using a tape and a protractor; the construction on paper gives an answer that can then be checked directly.

**A side and two angles** $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side  $BC$ largest angle  $\angle A$ **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH               | O'ZBEKCHA                             | РУССКИЙ                     |
|-----------------------|---------------------------------------|-----------------------------|
| Second criterion      | <b>Ikkinchi alomat</b>                | <b>Второй признак</b>       |
| Two angles and a side | <b>Ikki burchak va tomon</b>          | <b>Два угла и сторона</b>   |
| Adjacent side         | <b>Yondosh tomon</b>                  | <b>Прилежащая сторона</b>   |
| Angle sum             | <b>Burchaklar yig'indisi</b>          | <b>Сумма углов</b>          |
| Surveying             | <b>Yer o'lchash</b>                   | <b>Съёмка местности</b>     |
| Inaccessible          | <b>Yetib bo'lmaydigan</b>             | <b>Недоступный</b>          |
| Alternate angles      | <b>Ichki almashinuvchi burchaklar</b> | <b>Накрест лежащие углы</b> |
| Proof                 | <b>Isbot</b>                          | <b>Доказательство</b>       |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

The second criterion needs:

three sides

a side and two angles

two sides and an angle

three angles

The third angle:

must be given

follows from the sum

is irrelevant

is always  $90^\circ$

Must the side lie between the angles?

yes

no

only sometimes

it depends on the size

Three angles alone give:

congruence

similarity

nothing

equal areas

Parallel lines supply:

equal sides

equal angles

right angles

nothing

Surveying a river width uses:

the first criterion

the second criterion

the triangle inequality

Pythagoras

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS



1. The second criterion in letters

ANSWER ASA

2. A side of 7 with  $50^\circ$  and  $60^\circ$  on it: third angle

ANSWER  $70^\circ$

3. Two triangles with that data

ANSWER Congruent

4. Three equal angles give

ANSWER Similarity only

5. Alternate angles come from

ANSWER Parallel lines

6. Angles  $50^\circ$ ,  $70^\circ$ : the third

ANSWER  $60^\circ$

7. Is a side always needed?

ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. A side of 7,  $50^\circ$  on it and  $70^\circ$  opposite: congruent?

ANSWER Yes

2.  $AB \parallel CD$ ,  $AB = CD$ : which criterion?

ANSWER The second

3. What follows about  $AO$  and  $DO$ ?

ANSWER They are equal

4. Two triangles with  $\angle A = \angle D$ ,  $\angle B = \angle E$ ,  $AB = DE$

ANSWER Congruent

5. Two triangles with  $\angle A = \angle D$ ,  $\angle B = \angle E$  only

ANSWER Similar only

6. Angles  $40^\circ$ ,  $40^\circ$  with the included side 9

ANSWER Congruent to any such triangle

7. The third angle there

ANSWER  $100^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Prove that the diagonals of a parallelogram bisect each other

ANSWER ASA with alternate angles

2. Prove that a point on the bisector of an angle is equidistant from the arms

ANSWER ASA, then CPCT

3. Prove that the base angles of an isosceles triangle are equal, using ASA

ANSWER Compare the triangle with itself, reversed

4. A triangle with  $\angle A = 3x$ ,  $\angle B = 2x$ ,  $\angle C = x$

ANSWER  $90^\circ$ ,  $60^\circ$ ,  $30^\circ$

5. Why is ASA enough but SSA not?

ANSWER The third angle is forced; the third side is not

6. Two triangles share a side and have two pairs of equal angles

ANSWER Congruent

7. How many measurements are needed to fix a triangle?

ANSWER Three, including at least one length

## 07 Homework

### Homework — 5 tasks

Mark the equal angles and the equal side on the figure before writing the proof.

1. Two triangles have a side of 9 with angles  $45^\circ$  and  $65^\circ$  on it. Are they congruent?
2. Find the third angle in that triangle.
3. Prove that the diagonals of a parallelogram bisect each other.
4. Explain why three angles alone do not prove congruence.
5. Describe how to measure the width of a river using the second criterion.